

PARTS LIST

Semiconductors:

- IC1, IC2 - TEA2025 audio amplifier
- LED1-LED3 - 5mm LED
- BR1 - 3A, bridge rectifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 1-kilo-ohm
- R3-R6 - 10-ohm
- R7, R9 - 180-ohm
- R8, R10 - 20-ohm
- R11 - 470-ohm
- R12 - 2.2-kilo-ohm
- VR1-VR3 - 100-kilo-ohm potentiometer

Capacitors:

- C1-C3 - 330pF ceramic disk
- C4-C6 - 0.33 μ F ceramic disk
- C7-C13, C16, C19, C21 - 100 μ F, 25V electrolytic
- C14, C17, C26 - 1000 μ F, 25V electrolytic
- C15, C18, C20, C22 - 150nF ceramic disk
- C23-C25 - 100nF ceramic disk

Miscellaneous:

- LS1, LS2 - 4-ohm, 4W loudspeaker
- LS3 - 4-ohm, 6W loudspeaker
- HP1, HP2 - 8-ohm headphone
- CON1-CON8 - 2-pin connector
- CON9-CON10 - 2-pin terminal connector
- J1-J6 - Shorting jumper
- J7, J8 - Jumper wire
- 6-9V AC or 6-9V DC power supply
- 16-pin IC base for IC1 and IC2

(All these are available from kitsnspares.com)

responding capacitors C7 and C8 to stabilise.

The amplifier has six shorting jumpers J1 through J6 to have different modes of operation. Some of the configurations are given in Table 2.

Output power

The output power of the amplifier depends on the power supply and the load resistance (loudspeaker). Table 3 gives information about the approximate output powers (minimum, typical and maximum) of TEA2025 for different configurations and load resistances (R_{load}). IC1 works as two simple amplifiers and IC2 works as a single bridged amplifier with higher output power.

The input impedance of the IC TEA2025 is typically 30-kilo-ohm. Here we use three external potentiometers (VR1 through VR3), each of 100-kilo-ohm, to adjust the volume of the corresponding channel inputs.

The circuit has three LEDs. LED1 and LED2 are used as simple level indicators. These produce light pro-

portional to the level of the output signal of the third channel, which is the most powerful channel that works in bridged mode. LED3 is the power on/off indicator.

Construction and testing

An actual-size PCB layout for the amplifier is shown in Fig. 2 and its components layout in Fig. 3.

The circuit can be tested with DC input voltage (6V-9V) applied to connector CON9 or AC input voltage (6 to 9Vrms) applied to connector CON10. The test input signals (about 10mV rms, depending on the power supply) are applied to CON1, CON2 and CON3. **EFY**



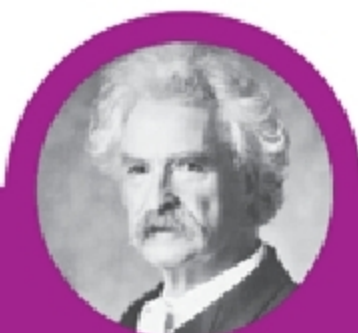
Petre Tzv Petrov was a researcher and assistant professor at the technical University of Sofia, Bulgaria, and expert-lecturer in OFPPT (Casablanca), Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria

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-Mark Twain

